INHERITANCE, ITERATORS, & GENERATORS Solutions

COMPUTER SCIENCE MENTORS 61A

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1 Inheritance

Inheritance lets a **subclass** reuse and extend the behavior of a **base class**.

```
class Animal:
    kind = 'animal'

    def __init__(self, name):
        self.name = name

    def introduce(self):
        return self.name + ' says ' + self.speak()

    def speak(self):
        return 'hello'

class Dog(Animal):
    kind = 'dog'

    def speak(self):
        return 'woof'
```

- **Attribute lookup:** Python looks up `obj.attr` by checking the instance first, then its class, then its base classes. Lookup does not search subclasses.
- **Inherited methods:** If a subclass does not define a method, it uses the method defined in its base class. For example, `Dog` inherits `__init__` and `introduce` from `Animal`.
- **Overriding:** A subclass can define its own version of an inherited method. Here, `Dog.speak` overrides `Animal.speak`.
- **The meaning of `self`:** When an inherited method runs, `self` is still the instance on which the method was called. Thus, `self.speak()` inside `Animal.introduce` uses `Dog.speak` when `self` is a `Dog`.
- **Assignment versus lookup:** `obj.attr = value` sets an instance attribute, whereas `Class.attr = value` sets a class attribute. Setting an instance attribute does not change the class attribute.
- **Calling a base-class method:** Inside a subclass method, **`super`**`().method(...)` can call the inherited implementation. For single inheritance, `Base.method(self, ...)` can also call it explicitly.

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Neither changes which object `self` refers to.

```
>>> d = Dog('Maple')
>>> d.name
'Maple'
>>> d.kind
'dog'
>>> d.introduce()
'Maple says woof'
>>> Animal.speak(d)
'hello'
>>> d.kind = 'friend'
>>> d.kind, Dog.kind
('friend', 'dog')
```

1. What would Python display in the land of Oz?

```
class Elphaba(object):
    magic = 'Defying Gravity'

    def __init__(self, magic):
        self.magic = magic

    def spell(self):
        return type(self).magic + ' and ' + self.magic

class Glinda(Elphaba):
    magic = 'Popular'

    def __init__(self, magic):
        Elphaba.__init__(self, 'good ' + magic)

elphaba = Elphaba('Wicked')
```

(a) `>>> Elphaba.magic`

`'Defying Gravity'`

(b) `>>> elphaba.magic`

`'Wicked'`

(c) `>>> elphaba.spell()`

`'Defying Gravity and Wicked'`

(d) `>>> Elphaba.spell()`

`Error`

(e) `>>> Elphaba.spell(elphaba)`

`'Defying Gravity and Wicked'`

(f)
```
>>> glinda = Glinda('Power')
>>> Glinda.magic
```

```
'Popular'
```

(g)
```
>>> glinda.magic
'good Power'
```

(h)
```
>>> glinda.spell()
'Popular and good Power'
```

2. What would Python display? Write the result of executing the following code and prompts. If nothing would happen, write "Nothing". If an error occurs, write "Error".

```
class ForceWielder():
    force = 25

    def __init__(self, name):
        self.name = name

    def train(self, other):
        other.force += self.force / 5

    def __str__(self):
        return self.name

class Jedi(ForceWielder):
    lightsaber = "blue"

    def __str__(self):
        return "Jedi " + self.name

    def __repr__(self):
        return f"Jedi({repr(self.name)})"

class Sith(ForceWielder):
    lightsaber = "red"
    num_sith = 0

    def __init__(self, name):
        super().__init__(name)
        Sith.num_sith += 1
        if self.num_sith != 2:
            print("Two there should be. No more, no less.")

    def __str__(self):
        return "Darth " + self.name

    def __repr__(self):
        return f"Sith({repr(self.name)})"
```

```
>>> anakin = Jedi("Anakin")
>>> anakin.lightsaber, anakin.force
```

```
('blue', 25)
```

```
>>> obiwan = Jedi('Obi-wan')
>>> anakin.master = obiwan
>>> anakin.master
```

```
Jedi("Obi-wan")
```

```
>>> Jedi.master
```

```
AttributeError
```

```
>>> obiwan.force += anakin.force
>>> obiwan.force, anakin.force
```

```
(50, 25)
```

```
>>> obiwan.train(anakin)
>>> obiwan.force, anakin.force
```

```
(50, 35.0)
```

```
>>> Jedi.train(obiwan, anakin)
>>> obiwan.force, anakin.force
```

```
(50, 45.0)
```

```
>>> sidious = Sith("Sidious")
```

```
Two there should be. No more, no less.
```

```
>>> ForceWielder.train(sidious, anakin)
>>> anakin.lightsaber = "red"
>>> anakin.lightsaber, anakin.force
```

```
('red', 50.0)
```

```
>>> Jedi.lightsaber
```

```
'blue'

>>> print(Sith("Vader"), Sith("Maul").num_sith)

Two there should be. No more, no less.
Darth Vader 3

>>> rey = ForceWielder("Rey")
>>> rey

<__main__.ForceWielder object>

>>> rey.lightsaber

AttributeError
```

3. Write `TeamBaller`, a subclass of `Baller`. An instance of `TeamBaller` cheers on the team every time it passes a ball/states it doesn't have the ball when it doesn't and returns a boolean of whether or not it passed the ball. If the `TeamBaller` did not pass the ball, it'll say it hasn't done so and return False. Assume `Baller` has a defined *pass_ball* method.

```
class Baller:
    all_players = []
    def __init__(self, name, has_ball = False):
        self.name = name
        self.has_ball = has_ball
        Baller.all_players.append(self)

    def pass_ball(self, other_player):
        if self.has_ball:
            self.has_ball = False
            other_player.has_ball = True
            return True
        else:
            return False

class BallHog(Baller):
    def pass_ball(self, other_player):
        return False


class TeamBaller(Baller):
    """
    >>> alyssa = BallHog('Alyssa')
    >>> cheerballer = TeamBaller('Esther', has_ball=True)
    >>> cheerballer.pass_ball(alyssa)
    Yay!
    True
    >>> cheerballer.pass_ball(alyssa)
    I don't have the ball
    False
    """
    def pass_ball(self, other):
        did_pass = Baller.pass_ball(self, other)
        if did_pass:
            print('Yay!')
        else:
            print("I don't have the ball")
        return did_pass
```

2 Iterators

An **iterable** is any object whose elements can be processed one at a time, such as a list, string, dictionary, or **range**. An **iterator** is an object that keeps track of a position in a sequence and can produce the next element on request.

- **iter**`(iterable)` returns an iterator over the elements of the iterable. For a list, each call creates a

new iterator; for an iterator, it returns the iterator itself.

- **next**(iterator) returns the next element and advances the iterator. When no elements remain, it raises StopIteration.

```
>>> s = [3, 4, 5]
>>> t = iter(s)
>>> next(t)
3
>>> next(t)
4
>>> u = iter(s)          # a separate iterator with its own position
>>> next(u)
3
>>> w = t                # not a new iterator: another name for the same one
>>> next(w)
5
>>> next(t)
StopIteration
```

Iterators only move forward, and calling **next** on any name for an iterator advances it for every name. A **for** loop calls **iter** on its iterable and repeatedly calls **next** until StopIteration.

Many built-in functions return iterators that compute their results **lazily**, meaning each element is computed only when **next** asks for it:

- **map**(func, iterable): iterate over func(x) for each x in the iterable
- **filter**(func, iterable): iterate over each x in the iterable for which func(x) is true
- **zip**(first_iter, second_iter): iterate over pairs (x, y)
- **reversed**(sequence): iterate over the sequence in reverse

Functions such as **list**, **sum**, **min**, **max**, **all**, and **any** consume an iterable. **all** stops at the first false value and **any** stops at the first true value, so they may not use up the whole iterator.

4. What is difference between an iterator and an iterable?

   Iterator: Mutable object that tracks a position in a sequence, advancing on each call to **next**

   Iterable: Represents a sequence and returns an iterator when passed to **iter**. For a list, each call creates a new iterator; an iterator is also an iterable, and **iter** on an iterator returns that same iterator.

   To use in an English sentence: Lists are "iterable". To go through a list, you make an object called an "iterator" to scan through the list.

5. What Would Python Display? If you think nothing will be displayed, write N/A.

```
>>> a = [1, 2, 3]
>>> x = iter(a)
```

N/A

```
>>> next(x)
```

1

```
>>> next(x)
```

2

```
>>> y = iter(a)
>>> next(y)
```

1

```
>>> next(x)
```

3

```
>>> next(x)
```

StopIteration Error

```
>>> z = iter(y)
>>> next(z)
```

2

```
>>> [next(y), next(y), next(z)]
```

StopIteration Error

```
>>> a = iter(filter(lambda x: x % 2, map(lambda x: x - 1, range(10))))
>>> next(a)
```

-1

```
>>> from functools import reduce
>>> reduce(lambda x, y: x + y, a)
```

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6. WWPD? Write what this python program will print.

```
a = [1, 'A', 'B', 'C', 5, 6, 7, 'D', 'E']
x = iter(a)
for i in range(5 - next(x)):
    next(x)
print(next(x))
y = iter(a)
print(next(y))
z = iter(y)
print(next(z))
```

```
6
1
A
```

7. Find an input to the year function that prints the following output: 2 0 2 6

```
def year(a):
    x = iter(a)
    y = iter(a)
    z = iter(x)
    for i in range(next(x)):
        y = iter(a)
        next(y)
    print(next(x))
    print(next(z))
    print(next(y))
    print(next(z))
```

```
year([__, 2, 0, 6, _____])
There can be any positive number in the first blank, and there can be any
   number of values after the 6.
```

3 Generators

A **generator function** is a function whose body contains **`yield`**. A normal function returns once; a generator function can yield many times. A **generator** is an iterator created automatically by calling a generator function.

```
>>> def plus_minus(x):
...     yield x
...     yield -x
>>> t = plus_minus(3)     # no body code runs yet
>>> t
<generator object plus_minus ...>
>>> next(t)               # run until the first yield
3
>>> next(t)               # resume after the first yield
```

```
-3
>>> next(t)                # the body finished
StopIteration
```

Calling a generator function does not run its body. The body runs only when **next** is called, and it pauses at each **yield**, keeping its local variables and its place in the code until the next call to **next**. When the body finishes (or reaches **return**), StopIteration is raised. Each call to a generator function creates an independent generator.

yield from <iterable> yields each element of the iterable in order, and is shorthand for a loop:

```
def a():
    yield from range(3)
```

```
def b():
    for x in range(3):
        yield x
```

Recursive generators combine **yield from** (to pass on everything the recursive call produces) with ordinary **yield** (to add values of their own).

8. Given the following code block, what is output by the lines that follow?

```
def foo():
    a = 0
    if a == 0:
        print("Hello")
        yield a
        print("World")
```

```
>>> foo()
```

```
<generator object>
```

```
>>> foo_gen = foo()
>>> next(foo_gen)
```

```
Hello
0
```

```
>>> next(foo_gen)
```

```
World
StopIteration
```

```
>>> for i in foo():
...     print(i)
```

```
Hello
0
World
```

9. Write a generator function that takes in an iterator `it` and yields the running total of the elements produced by `it`.

```
def accumulate(it):
    """
    >>> def all_ints():
    ...     i = 0
    ...     while True:
    ...         yield i
    ...         i += 1
    >>> a = accumulate(all_ints())
    >>> [next(a) for x in range(6)]
    [0, 1, 3, 6, 10, 15]
    >>> list(accumulate(iter([1, 2, 3])))
    [1, 3, 6]
    >>> list(accumulate(iter([])))
    []
    """


def accumulate(it):
    total = 0
    for value in it:
        total += value
        yield total
```

10. Write a generator that, given m (the amount of money you have), pc (the cost of one pear), and ac (the cost of one apple) yields all possible combinations of fruit that you can buy such that the money left over cannot buy another fruit. Assume `m` is nonnegative and both prices are strictly positive. Combinations of the same number of each fruit in different orders is okay. It is also okay if each combination has an extra space at the end.

```
def fruitOptions(m, pc, ac):
    """
    >>> print(list(fruitOptions(10, 2, 5)))
    ['pear pear pear pear pear ', 'pear pear apple ', 'pear apple pear
       ',
     'apple pear pear ', 'apple apple ']
    """
    if ____________________________________:
        yield ""
    if m >= pc:
        for ____________________________________________:
            ____________________________________
    if m >= ac:
        for ____________________________________________:
            ____________________________________________


    def fruitOptions(m, pc, ac):
        if m < pc and m < ac:
            yield ""
        if m >= pc:
            for p in fruitOptions(m-pc, pc, ac):
                yield "pear " + p;
        if m >= ac:
            for a in fruitOptions(m-ac, pc, ac):
                yield "apple " + a;
```